COOLSHADE · DESIGN DECISION RECORD

CoolShade recommendations accepted

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CSH-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/coolshade		
Source	docs/decisions/0002-recommendations-accepted.md @ ae091ec		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); record every newly decided item, what changed in the repo and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Amish wrote, in chat on 2026-09-25: "i accept all your recommendations, go with them across all repos."

Context

After the TRL 3 session, `docs/REVIEW.md` and CSH-DDR-001 listed seven design choices adopted for TRL 3 and open for Amish's review (D1 to D7), the budget (O1), the first site and partner (O2), and three new items raised by the TRL 3 calculations (R10 wind with the fabric fitted, R4 air cooling and fans, R8 energy margin). Every item with a recommendation is now decided as recommended. Where several options were offered, the recommended option is the decision. Items without a recommendation stay open. TRL 4 stays on hold.

Options considered

The options for each item are those listed in `docs/REVIEW.md` (sessions 2026-09-25, /populate and TRL 3) and in CSH-DDR-001.

Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
D1	Misting pressure class	Low pressure, about 7 bar, 12 V, off-grid	Status only; design already used it
D2	Activation thresholds	Fixed: 32 °C or more, 60 % RH or less, presence in the last 2 min, tank above low level	Status only (CSH-REQ-001 R6, CSH-PRC-001)
D3	Fans	None; confirmed by the R4 decision below	Status only
D4	Water supply	Tank as standard; mains with float valve and backflow preventer optional where rules allow	Status only
D5	Canopy size and structure	3.0 x 2.4 m, four 80 x 80 x 3 mm posts with knee braces, bolted to an existing slab	Status only
D6	Storm practice	Removable fabric	Now tied to the restated R10 below
D7	Status link	None for the first prototype	Status only
O1	Budget	Option (a) of the TRL 3 review: `budget_usd` $775 (supersedes the TRL 2 recommendation of $750)	`project.yaml` `budget_usd` 700 to 775; README, CSH-PRB-001, CSH-REQ-001 R13, CSH-PRC-001, CSH-CAL-001 section J, BOM notes, concept sheet
N1	R10 fabric-on wind case	Option (a): restate R10 as "frame and anchors survive 30 m/s with the fabric removed; fabric fitted only in forecast gusts below 15 m/s (54 km/h)"	CSH-REQ-001 R10 restated; CSH-CAL-001 new lines G13 (fabric off, 30 m/s: posts 0.14, purlins 0.15, anchors about 2.8 kN) and G14 (fabric on, 15 m/s: long beams 0.88); R10 moves from not met to met on paper; GA drawing notes; BOM line 3 note; safety text
N1c	Load-release cord lacing	Option (c), to be studied at TRL 4	Decided but on hold: TRL 4 is on hold by Amish's instruction. Nothing built or tested
N2	R4 and fans	Option (a): relax R4 to "2 °C while spraying"; no fans	CSH-REQ-001 R4 restated; CSH-CAL-001 E3 (2.71 °C while spraying at 1 m/s, held to 1.35 m/s); R4 moves from not met to met
N3	R8 energy margin	Option (a): 25 Ah battery instead of 20 Ah; keep the fail-safe normally open drain valve	BOM line 6: 12.8 V 25 Ah, $85 to $100; `cad/src/model.py` battery 80 to 100 mm wide; STEP and STL re-exported; CSH-CAL-001 F2 (1.005 to 1.256 design days); R8 moves from at risk to met; GA drawing CSH-DWG-001 Rev P1 to P2; media re-rendered

Table 2. Items still open.

#	Item	Status
O2	First site type (bus stop, market or queue) and co-design partner. No recommendation was made.	Proposed, awaiting Amish

The suggestion in the TRL 3 review (a continuous spray at half the flow instead of 20 s pulses) was listed as a suggestion only, not a recommendation on a decision item, so it is not applied.

Consequences

- Parts cost: $758.00 to $773.00, against the new $775 budget, $2.00 (0.3 %) under. R13 moves from not met to at risk, because the steel prices look low until quoted.
- Requirement status (CSH-CAL-001 v0.2): none not met (was 3: R4, R10, R13); 3 at risk (R9, R13, R16; was R8, R9, R16); 1 not verifiable at TRL 3 (R5); 13 met (was 10).
- Documents bumped: CSH-PRB-001 v0.4, CSH-PRC-001 v0.4, CSH-REQ-001 v0.4, CSH-CAL-001 v0.2, CSH-DDR-001 v0.2, drawing CSH-DWG-001 Rev P2. No pitch or problem rewording was recommended, so `project.yaml` pitch and problem are unchanged.
- Cross-repo actions: none. No FieldNode link is fitted (D7), so CoolShade places no demand on another repo.
- `trl: 3` and `trl_target: 3` are unchanged. TRL 4 is on hold by Amish's instruction; nothing in this record authorizes building, testing, purchasing, PCB or firmware work.